

Session Objectives

1. Describe the classifications, epidemiology, and risk factors associated with endocarditis.
2. Identify common and atypical bacterial and fungal causes of endocarditis.
3. Describe the pathogenesis associated with different microbial causes of endocarditis.
4. Identify unique microbial features and virulence factors associated with the different pathogens.
5. Outline the laboratory techniques used for the identification and diagnosis of endocarditis.

Definitions

Infective Endocarditis (IE) – **Infection** and inflammation of the heart's **endocardium**, specifically the **valve(s)** of the heart

Classifications

By Onset:

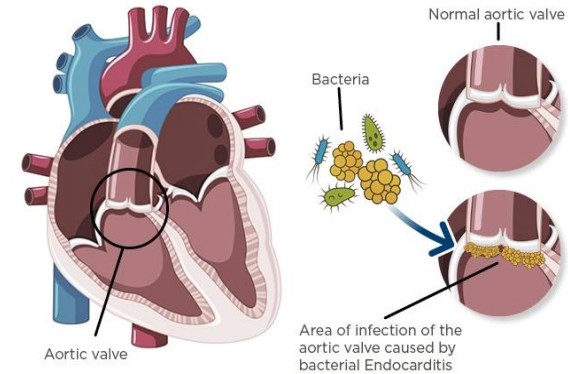
- **Acute IE** – Rapid progression, highly virulent organisms
 - Markedly febrile, lasting days - weeks
- **Subacute IE** – Slow progression, less virulent organisms,
 - Lower fevers, can last for several weeks

By Source:

- Native Valve Endocarditis (**NVE**)
- Prosthetic Valve Endocarditis (**PVE**)
- **IV Drug use-associated IE** – often right sided (Tricuspid valve)

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Risk Factors

- Age >60
- Injection drug use
- Poor dentition/dental infection
- History of infective endocarditis
- Structural heart disease (valvular, congenital)
- Chronic hemodialysis
- Procedures (intravascular device, cardiac implantable electronic device, cardiac bypass surgery)

Epidemiology

- Incidence = 15/100,000 in the US
- More than 50% patients are over the age of 60
- 3x as common in males vs females
- Bacteria are by far the most common cause of endocarditis (fungi rare)

Microbial Causes of IE

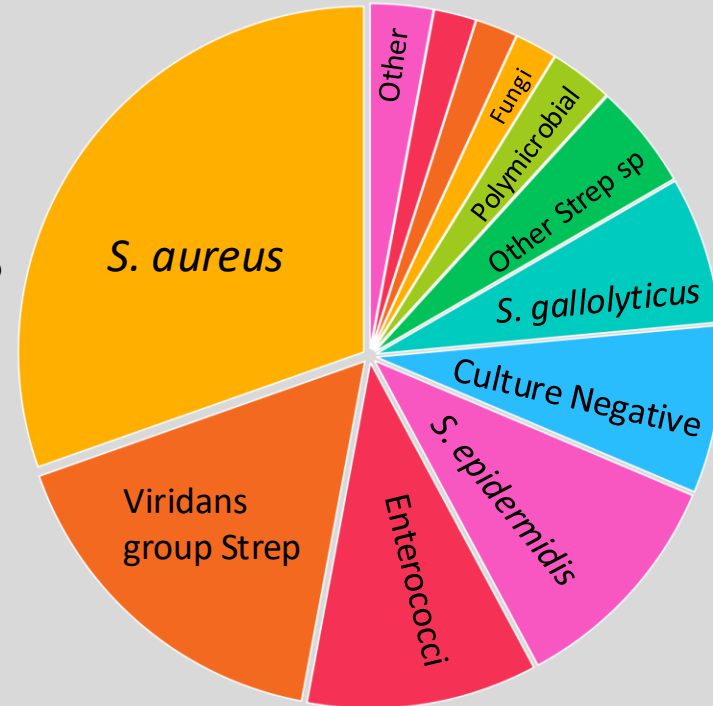
Three **most common causes** of IE worldwide: Staphylococci, Streptococci and enterococci

The distribution of pathogens are as follows:

1. *S. aureus* – 30% of cases
2. Viridans group Streptococci – 16%
3. Enterococci – 11%
4. Coagulase-negative Staphylococci (*S. epidermidis*) – 11%
5. Culture negative endocarditis – 8%
6. *Streptococcus gallolyticus* (Formerly *S. bovis*) – 7%
7. Other streptococci – 5%
8. Polymicrobial – 3%
9. Fungi – 2%
10. Non-HACEK gram-negative bacteria – 2%
11. HACEK Organisms – 2%
12. Other - 3%

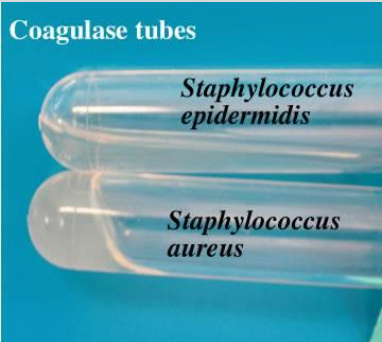
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Staphylococci – aureus and epidermidis

- All Staph spp. are Gram +, **catalase +**, growing in **clumps**
- *S. aureus* is coagulase +
- *S. epidermidis* is novobiocin sensitive



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Feature	<i>S. aureus</i>	<i>S. epidermidis</i>
Gram Stain and Morphology	Gram + Cocci	Gram + Cocci
Coagulase	POSITIVE – clots plasma	NEGATIVE
Biofilm	Rare (not typical)	Strong Biofilm formation , especially on prosthetic devices
Hemolysis	Beta hemolytic (complete)	Gamma Hemolytic (non-hemolytic)
Typical Disease	<u>MCC of Acute</u> Endocarditis (esp. IV Drug users)	<u>Subacute</u> Prosthetic valve endocarditis

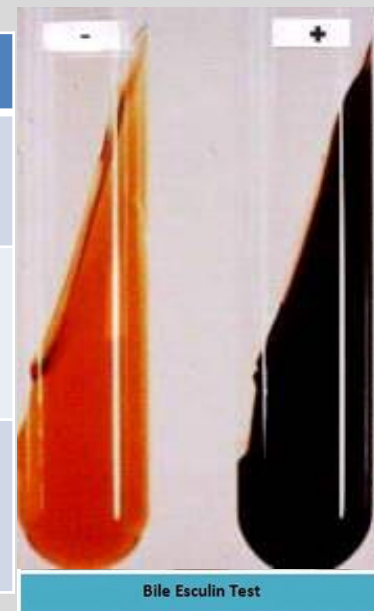
Streptococci – *S. viridans*, *gallolyticus* and *E. faecalis/faecium*

- Catalase test **essential first step** to differentiate Staph vs Strep
- Viridans - Subacute IE from **Dental procedures**
 - Dextran-mediated** adherence to fibrin-platelet aggregates
- Gallolyticus (Bovis) – screen for **colon cancer**

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Species	Lancefield Group	Key Features	Diseases
<i>S. viridans</i> (<i>S. sanguinis</i> , <i>S. mutans</i> , <i>S. mitis</i>)	none	Oral flora	MCC of subacute IE (esp. after dental procedures)
<i>S. gallolyticus</i> (FKA <i>S. Bovis</i>) ~ Bovis in Blood, Cancer in Colon ~	Group D (non-enterococcus)	- Bile esculin + Does not grow in 6.5% NaCl - GI tract	STRONG ASSOCIATION with Colon Ca and GI neoplasms
<i>Enterococcus faecalis/faecium</i>	Group D (enterococcus)	- Bile Esculin + PYR + Grows in 6.5% NaCl - GI/GU tract	Subacute IE after GU/GI procedures



Atypical Gram-Positive Bacteria – NVS, Corynebacteria, Listeria

- Nutritionally Variant Strep (**NVS**) – Oral flora
 - **B6 supplemented media**, satellitism around *S. aureus*
- **Corynebacterium** (diphtheroids) – pleomorphic rods (club shaped) in patients with **prosthetic devices**, Isolated on **Loeffler's medium**
- **Listeria** – elderly/pregnant/immunocompromised patient, **Foodborne** (cold enrichment)

Atypical Gram-Negative Bacteria – HACEK Organisms

- **HACEK** organisms: **MCC of gram-negative endocarditis** in patients that **do not abuse IV drugs**
- All are bacilli found in **normal oral flora**, slow-growing
- Consider **dental disease/oral trauma** as portals of entry

Organism
H aemophilus sp*
A ctinobacillus sp*
C ardiobacterium hominis
E ikenella corrodens
K ingella kingae

* = Haemophilus and Actinobacillus were reclassified into Aggregatibacter Genus

Virulence Factors Exacerbating Infective Endocarditis

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- 1) **Immune Resistance** – IE causing bacteria can resist complement mediated lysis or phagocytosis
 - **Capsule** protection is a major defense mechanism

- 2) **Adherence to Damaged Valves or Vegetations**
 - **Dextran-Mediated** adherence - extracellular polysaccharide (strep)
 - **FimA protein** - Bacterial surface adhesin (strep)
 - **Fibronectin Binding proteins** - Bacterial surface adhesin (*S. aureus*)

Bacterial Entry, Spread, and Multiplication

Bacteria/fungi commonly enter bloodstream via:

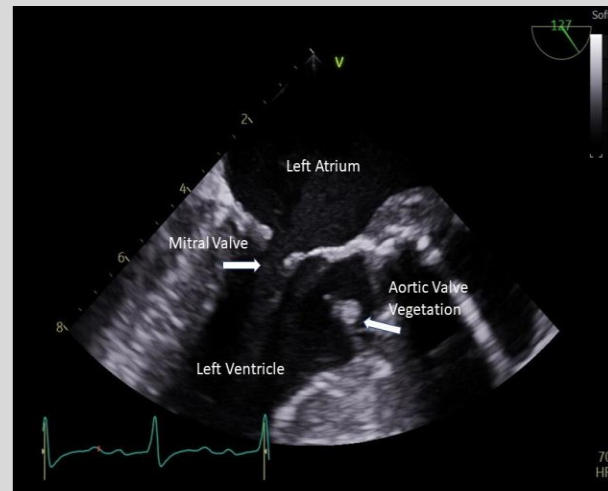
- Trauma to heavily colonized mucosa (dental procedures)
- Indwelling vascular access
- IV Drug use
 - Right sided endocarditis (Tricuspid abnormalities)
 - ***Staph aureus*, *Candida*, *Pseudomonas*** most common

Vegetations:

- Damaged valves become “sticky” from **fibrin and thrombin deposition**, increasing surface area and allows for the deposition of infective organisms, complement and antibodies
- Mitral Valve (Most common overall) > Aortic Valve >> Tricuspid Valve >>>>> Pulmonic valve

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Question 3:

A 68-year-old woman presents to her family physician with the signs and symptoms of cardiovascular abnormalities. A heart murmur was heard on auscultation, and valvular vegetations were revealed by TTE. Cultures done on blood agar identified the pathogen as being a Gram-Positive cocci growing in clusters that is catalase positive and novobiocin sensitive. Which of the following is the likely cause of this patient's condition?

- A. *Enterococcus faecalis*
- B. *Chlamydophila pneumoniae*
- C. *Staphylococcus epidermidis*
- D. *Bartonella henselae*
- E. *Mycobacterium tuberculosis*

Question 4:

A 66-year-old man is diagnosed clinically with subacute infective endocarditis. Routine blood cultures failed to grow out any organisms until the 10th day of incubation. A gram stain of the culture revealed pleomorphic bacilli that are part of the normal mouth flora and cause <5% of endocarditis cases. The likely pathogen, that is causing this man's disease, is:

- A. *Candida albicans*
- B. One of the HACEK bacteria
- C. Viridans group of *Streptococci*
- D. Coagulase-negative *Staphylococci*
- E. Coagulase-positive *Staphylococci*

Summary

- Various types of microbes (bacterial & fungal) are capable of causing infective endocarditis (IE).
- The major causes include the Staph. and Strep. families, but other less common microbes should be considered.
- The major disease manifestation is a vegetation that typically forms on one of the heart valves.
- If not diagnosed early or if left untreated, IE could lead to serious systemic complications.